Uniform Logarithmic Gamma Coercivity at Fixed Exponential Complexity

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Abstract

Let W be a positive C^1 weight such that W and W' decrease faster than every exponential, and let P range over exponential polynomials of fixed total confluent complexity whose frequencies lie in a fixed horizontal strip. We prove a uniform comparison between the weighted norm of WP in $L^2(\mathbb{R}, dx/W)$ and its ordinary L^2 norm, with no separation assumption on the frequencies. When every real frequency tends in absolute value to infinity, the normalized Fourier mass of WP escapes every fixed compact interval, uniformly over the same class. A compact family of constant-coefficient ordinary differential equations controls collisions; a finite-cluster argument handles arbitrary real separations. Consequently the exact logarithmic Gamma form, minus any bounded self-adjoint perturbation in the weighted space, has Rayleigh quotient tending uniformly to $+\infty$. We verify the hypotheses for the classical theta weight and prove directly that its complete prime-power correction is norm-absolutely summable. An entire two-frequency formula and confluent cosine consequences are included. Auxiliary examples show why complexity cannot simply be omitted; they are not actual zeta-zero jets and give no negative-form witness. No complexity-independent threshold or assertion about all zeta-zero jets is proved.

Keywords. Exponential polynomials; confluent frequencies; weighted norms; Fourier escape; logarithmic Gamma form; theta weight.

1 Introduction and main theorem

High-frequency modulation makes a logarithmically growing Fourier multiplier large on any one fixed envelope. Uniformity is subtler when the envelope is allowed to vary with several complex frequencies, colliding frequencies, and polynomial factors. The coefficients of a nearly colliding exponential basis can diverge even when the represented function remains small. Conversely, a collection of frequencies can split into widely separated groups, so no one bounded frequency window contains the whole configuration after a common modulation.

We address these two issues at fixed total complexity. The local estimate uses the complete annihilating differential equation rather than the coefficients in an exponential basis. The global argument extracts finitely many frequency clusters and controls their cross inner products before normalizing individual envelopes. The resulting constants are uniform in the real-frequency configuration, but the proof is nonquantitative and keeps the complexity fixed throughout.

All spaces are complex, and inner products are linear in the first variable. Ordinary Lebesgue $L^2(\mathbb{R})$ norms are denoted by $\|\cdot\|_2$. We use the unitary Fourier transform

$$\widehat{u}(t) = (2\pi)^{-1/2} \int_{\mathbb{R}} u(x) e^{-itx}\, dx. \tag{1}$$

Definition 1.1 (Admissible weight)**.** A function $W : \mathbb{R} \to (0, \infty)$ is admissible if it is C^1 and, for every $a > 0$, there is $C_a < \infty$ such that

$$W(x) + |W'(x)| \le C_a e^{-a|x|} \qquad (x \in \mathbb{R}). \tag{2}$$

Set $\mathcal{H}_W = L^2(\mathbb{R}, dx/W)$, with $\|u\|_W^2 = \int_{\mathbb{R}} |u(x)|^2/W(x)\,dx$.

The assumptions imply that W is bounded, has a positive minimum on every compact interval, and tends to zero at both ends. In particular,

$$\|u\|_2^2 \le \|W\|_\infty \|u\|_W^2 \qquad (u \in \mathcal{H}_W). \tag{3}$$

The converse below is specific to a nonlinear class, not to all of $\mathcal{H}_W$.

Definition 1.2 (Exponential complexity)**.** For $N \in \mathbb{N}$ and $\beta \ge 0$, let $\mathcal{E}_{N,\beta}$ consist of zero and all functions

$$P(x) = \sum_{j=1}^{r} p_j(x) e^{i\lambda_j x}, \qquad \sum_{j=1}^{r} (\deg p_j + 1) \le N, \qquad |\operatorname{Im} \lambda_j| \le \beta, \tag{4}$$

where the $\lambda_j \in \mathbb{C}$ are distinct and the p_j are nonzero polynomials. Repeated frequencies are merged first. Let $\mathcal{E}_{N,\beta}(T)$ be the subclass for which $|\operatorname{Re} \lambda_j| \ge T$ for every displayed frequency.

The cost of a degree-k polynomial at one frequency is $k + 1$, regardless of how many of its coefficients vanish. Distinct exponential-polynomial terms are linearly independent, as recalled in Section 7, so the merged nonzero representation is unique. These classes need not be closed under addition at fixed N.

For $s > 0$ define

$$\rho(s) = \frac{e^{-s/2}}{1 - e^{-2s}}, \qquad \mathfrak{t}_\Gamma[u] = \int_0^\infty \rho(s) \|u(\cdot + s) - u\|_2^2 \, ds, \qquad \mathcal{X}_W = \{u \in \mathcal{H}_W : \mathfrak{t}_\Gamma[u] < \infty\}. \tag{5}$$

The nonnegative integral is interpreted without cancellation. If D is any bounded self-adjoint operator on $\mathcal{H}_W$, set

$$\mathfrak{q}_D(u, v) = \mathfrak{t}_\Gamma(u, v) - \langle Du, v\rangle_W, \qquad \mathfrak{q}_D[u] = \mathfrak{q}_D(u, u). \tag{6}$$

Here $\mathfrak{t}_\Gamma(u, v)$ is the polarized difference form. Every WP in Definition 1.2 belongs to $\mathcal{X}_W$, as proved below.

Theorem 1.3 (Uniform finite-complexity coercivity)**.** *Fix an admissible weight W, $N \in \mathbb{N}$, and $\beta \ge 0$.*

(i) *There is $C = C(N, \beta, W) < \infty$ such that*

$$\|WP\|_W^2 \le C \|WP\|_2^2 \qquad (P \in \mathcal{E}_{N,\beta}). \tag{7}$$

(ii) *For every fixed $0 \le R < \infty$,*

$$\varepsilon_{N,\beta,W}(T; R) := \sup_{0 \ne P \in \mathcal{E}_{N,\beta}(T)} \frac{\|\mathbf{1}_{[-R,R]} \widehat{WP}\|_2}{\|WP\|_2} \longrightarrow 0 \qquad (T \to \infty). \tag{8}$$

(iii) *For every bounded self-adjoint D on $\mathcal{H}_W$,*

$$\inf_{0 \ne P \in \mathcal{E}_{N,\beta}(T)} \frac{\mathfrak{q}_D[WP]}{\|WP\|_2^2} \longrightarrow +\infty, \qquad \inf_{0 \ne P \in \mathcal{E}_{N,\beta}(T)} \frac{\mathfrak{q}_D[WP]}{\|WP\|_W^2} \longrightarrow +\infty. \tag{9}$$

No lower bound on frequency separation or bound on coefficient conditioning is required. The constants and height thresholds may depend on N, β, W, and, in (iii), $\|D\|$.

In particular, for every prescribed $L > 0$ there is $T_L < \infty$ such that $\mathfrak{q}_D[WP] \geq L\|WP\|_W^2$ for every $P \in \mathcal{E}_{N,\beta}(T_L)$. This is uniform in the represented function norm, not in the Euclidean norm of raw exponential coefficients. The theorem gives no effective value of T_L and no uniformity as $N \to \infty$.

Section 2 fixes the exact Gamma form and its domain. Sections 3–5 prove the theorem. Section 6 verifies the theta and full-prime specialization without importing an energy-realization theorem. The remaining sections record finite cosine-jet consequences and the limitations of the complexity hypothesis. No historical-priority claim is made.

2 The exact Gamma form

Lemma 2.1 (Multiplier and domain)**.** *For $u \in \mathcal{H}_W$,*

$$\mathfrak{t}_\Gamma[u] = \int_{\mathbb{R}} d_\Gamma(t)|\widehat{u}(t)|^2\,dt, \qquad d_\Gamma(t) = 2\int_0^\infty \rho(s)(1-\cos(ts))\,ds. \tag{10}$$

The multiplier is nonnegative and satisfies

$$d_\Gamma(t) \geq \frac{1}{2}\log(1+4t^2). \tag{11}$$

Moreover, for a constant $C_\Gamma < \infty$,

$$\mathfrak{t}_\Gamma[u] \leq C_\Gamma(\|u\|_2^2 + \|u'\|_2^2) \qquad (u \in H^1(\mathbb{R})). \tag{12}$$

Thus $WP \in \mathcal{H}_W \cap H^1(\mathbb{R}) \subset \mathcal{X}_W$ for every exponential polynomial P.

Proof. By (3), $u \in L^2$. Plancherel for each shift and Tonelli for the nonnegative integrand give (10), including when either side is infinite. Near zero, $\rho(s) = O(s^{-1})$; at infinity, $\rho(s) = O(e^{-s/2})$. Thus the integral defining $d_\Gamma(t)$ converges for each t. The estimates

$$\|u(\cdot+s)-u\|_2 \leq s\|u'\|_2, \qquad \|u(\cdot+s)-u\|_2 \leq 2\|u\|_2$$

prove (12) by splitting at $s = 1$.

Since $1 - e^{-2s} \leq 2s$,

$$d_\Gamma(t) \geq I(t) := \int_0^\infty \frac{e^{-s/2}(1-\cos(ts))}{s}\,ds.$$

Differentiation under the integral on compact t intervals gives $I'(t) = \int_0^\infty e^{-s/2}\sin(ts)\,ds = t/(t^2 + 1/4)$. Integrating from 0 yields $I(t) = \frac{1}{2}\log(1+4t^2)$. Finally P and P' grow at most like a polynomial times an exponential. Equation (2) makes $\int W|P|^2$, $\int W^2|P|^2$, and $\int |W'P + WP'|^2$ finite. This proves admission. □

For reference, the exact digamma normalization is

$$d_\Gamma(t) = \operatorname{Re}\psi\left(\frac{1}{4} + \frac{it}{2}\right) - \psi(1/4), \qquad \psi = \Gamma'/\Gamma. \tag{13}$$

Indeed, subtract the integral representations

$$\psi(z) + \gamma = \int_0^\infty \frac{e^{-r} - e^{-zr}}{1-e^{-r}}\,dr \qquad (\operatorname{Re} z > 0)$$

at $z = 1/4 + it/2$ and $z = 1/4$, take real parts, and set $r = 2s$ [1, Eq. 5.9.16]. This identity names the form; only the directly proved bound (11) is needed for coercivity.

Proposition 2.2 (Closed form)**.** *The space $\mathcal{X}_W$, with norm $\|u\|_X^2 = \|u\|_W^2 + \mathfrak{t}_\Gamma[u]$, is complete. The form $\mathfrak{t}_\Gamma$ is densely defined and nonnegative on $\mathcal{H}_W$. The form $\mathfrak{q}_D$ is closed and bounded below by $-\|D\|\|u\|_W^2$.*

Proof. If u_n is Cauchy in $\|\cdot\|_X$, it converges to u in $\mathcal{H}_W$, hence in ordinary L^2. In Fourier variables, $d_\Gamma^{1/2}\widehat{u}_n$ is Cauchy in L^2. A subsequence of the two convergent L^2 sequences converges almost everywhere, which identifies the second limit as $d_\Gamma^{1/2}\widehat{u}$. Therefore $u \in \mathcal{X}_W$ and $u_n \to u$ in $\|\cdot\|_X$. Compactly supported smooth functions lie in $\mathcal{X}_W$ by Lemma 2.1 and are dense in $\mathcal{H}_W$: truncate first, then use the positive continuous weight bounds on a compact interval and ordinary smooth approximation. Finally $|\langle Du, u\rangle_W| \le \|D\|\|u\|_W^2$ and

$$\|u\|_X^2 \le \mathfrak{q}_D[u] + (\|D\| + 1)\|u\|_W^2 \le \mathfrak{t}_\Gamma[u] + (2\|D\| + 1)\|u\|_W^2.$$

The shifted form norm is equivalent to the complete X norm. □

No ordinary Fourier projection is asserted to preserve $\mathcal{H}_W$. Fourier cutoffs below are used only to estimate integrals in ordinary L^2.

3 Uniform envelopes in a bounded frequency window

Lemma 3.1 (Compact ODE estimate)**.** *Fix $N \in \mathbb{N}$ and $0 \le M < \infty$. If P has complexity at most N and all its frequencies obey $|\lambda_j| \le M$, then, with $A = WP$,*

$$\|A\|_W + \|A'\|_2 \le C_{N,M,W}\|A\|_2. \tag{14}$$

The estimate is uniform when distinct frequencies approach one another and when polynomial coefficients vary arbitrarily.

Proof. Fix the total order $m = \sum_j(\deg p_j + 1) \le N$. The function P satisfies the monic differential equation

$$\prod_j(\partial_x - i\lambda_j)^{\deg p_j + 1}P = 0. \tag{15}$$

The coefficients of the monic polynomial on the left lie in a compact set: they are continuous symmetric polynomials in m roots from a closed disk. Repeated roots are included in that set. Let B_a be the corresponding companion matrix and set $Y = (P, P', \dots, P^{(m-1)})^{\mathsf{T}}$. Then $Y' = B_aY$, and $\|B_a\| \le L$ uniformly in this compact family.

We claim the uniform observability estimate

$$|Y(0)|^2 \le C_m \int_{-1}^{1} |P(x)|^2\,dx. \tag{16}$$

Otherwise there would be coefficient vectors a_n in the compact family and unit initial vectors v_n for which the integrals tend to zero. Extract $a_n \to a$ and $v_n \to v$, with $|v| = 1$. The power series of the matrix exponential converges uniformly for $|x| \le 1$ and bounded matrices, so the first components of $e^{xB_{a_n}}v_n$ converge uniformly to that of $e^{xB_a}v$. The latter component vanishes identically on $[-1, 1]$. Its derivatives of orders $0, \dots, m-1$ at zero are the components of v, so $v = 0$, a contradiction.

The matrix exponential estimate gives $|Y(x)| \le e^{L|x|}|Y(0)|$ for all real x. The same type of bound holds for P', including when $m = 1$ by its differential equation. Since $w_* := \min_{[-1,1]} W > 0$,

$$\int_{-1}^{1} |P|^2\,dx \le w_*^{-2}\|WP\|_2^2.$$

Combining this with (16) yields

$$|P(x)| + |P'(x)| \le Ce^{L|x|}\|WP\|_2. \tag{17}$$

The integrals of $e^{2L|x|}W$, $e^{2L|x|}W^2$, and $e^{2L|x|}|W'|^2$ are finite by (2). Integrating (17) and using $(WP)' = W'P + WP'$ proves the lemma for order m. Take the maximum of the finitely many constants for $1 \le m \le N$. □

The compact family is formed only after frequencies are placed in a bounded window. The companion matrices of an original configuration with unbounded real frequencies need not be bounded. No inverse Vandermonde matrix or lower bound on root spacing is used.

4 Finite clusters at arbitrary real separations

Lemma 4.1 (Subsequential cluster decomposition)**.** *Let $P_n \in \mathcal{E}_{N,\beta}$ be nonzero. There is a subsequence, still indexed by n, and a fixed finite collection of clusters $\mathcal{C}$ such that*

$$WP_n = \sum_{C\in\mathcal{C}} e^{it_{C,n}x}A_{C,n}, \qquad A_{C,n} = WQ_{C,n}, \tag{18}$$

where each $t_{C,n}$ is the real part of an original frequency, all frequencies of each $Q_{C,n}$ lie in a fixed bounded disk depending on C, and $|t_{C,n} - t_{D,n}| \to \infty$ for $C \ne D$. There are constants $L_C < \infty$, independent of n, such that

$$\|A_{C,n}\|_W + \|A'_{C,n}\|_2 \le L_C\|A_{C,n}\|_2. \tag{19}$$

Moreover,

$$\|WP_n\|_2^2 = (1 + o(1)) \sum_{C\in\mathcal{C}} \|A_{C,n}\|_2^2, \tag{20}$$

where the relative error is independent of the cluster amplitudes.

Proof. Extract a subsequence on which the number of distinct frequencies and all polynomial degrees are fixed; there are only finitely many such choices. Label the frequencies along this subsequence. For every pair j, k, extract further so that $\operatorname{Re}\lambda_{j,n} - \operatorname{Re}\lambda_{k,n}$ converges in the extended real line. There are finitely many pairs. Declare $j \sim k$ when this limit is finite. Reflexivity and symmetry are immediate; transitivity follows by adding the two convergent real differences. The equivalence classes are the clusters.

Choose a member j_C in each cluster and put

$$t_{C,n} = \operatorname{Re}\lambda_{j_C,n}, \qquad Q_{C,n}(x) = \sum_{j\in C} p_{j,n}(x)e^{i(\lambda_{j,n}-t_{C,n})x}.$$

Within a cluster the real offsets are bounded, while the imaginary parts remain in $[-\beta, \beta]$. Between distinct clusters the carrier differences tend in modulus to infinity. This proves the stated decomposition and, by Lemma 3.1, (19).

For distinct clusters, integration by parts gives

$$\begin{aligned}\left|\langle e^{it_{C,n}x}A_{C,n}, e^{it_{D,n}x}A_{D,n}\rangle_2\right| &\le \frac{\|A'_{C,n}\|_2\|A_{D,n}\|_2 + \|A_{C,n}\|_2\|A'_{D,n}\|_2}{|t_{C,n} - t_{D,n}|} \\ &\le \frac{L_C + L_D}{|t_{C,n} - t_{D,n}|}\|A_{C,n}\|_2\|A_{D,n}\|_2. \end{aligned} \tag{21}$$

The product $A_{C,n}\overline{A_{D,n}}$ belongs to $W^{1,1}(\mathbb{R})$ by Cauchy–Schwarz. Its endpoint limits are zero: the derivative is integrable, so the limits exist, and its integrability forces those limits to vanish. Thus the integration by parts is valid.

Write $s = |\mathcal{C}| \le N$ and $h_{C,n} = \|A_{C,n}\|_2$. If $s > 1$, let δ_n be the largest coefficient on the right of (21). Then $\delta_n \to 0$ and

$$\left| \|WP_n\|_2^2 - \sum_C h_{C,n}^2 \right| \le 2\delta_n \sum_{C<D} h_{C,n} h_{D,n} \le (s-1)\delta_n \sum_C h_{C,n}^2.$$

For $s = 1$ the identity is exact. This proves (20); in particular, for all sufficiently large n,

$$\sum_C \|A_{C,n}\|_2^2 \le 2\|WP_n\|_2^2. \tag{22}$$

□

The constants L_C depend on the bounded offset windows found along the subsequence. They are not yet universal constants for the original class. This is sufficient for the contradiction arguments that follow. Crucially, (22) is established without any prior bound on the separate envelope norms or on the raw coefficients.

5 Proof of uniform comparison and coercivity

Proof of Theorem 1.3(i). Suppose no constant in (7) existed. Choose nonzero $P_n \in \mathcal{E}_{N,\beta}$ with $\|WP_n\|_W^2 > n\|WP_n\|_2^2$. Apply Lemma 4.1. Multiplication by a real modulation is an isometry in both weighted and ordinary norms. If $s = |\mathcal{C}|$ and $L = \max_C L_C$, the triangle inequality and Cauchy–Schwarz give

$$\|WP_n\|_W^2 \le s \sum_C \|A_{C,n}\|_W^2 \le sL^2 \sum_C \|A_{C,n}\|_2^2 \le 2sL^2 \|WP_n\|_2^2$$

for all sufficiently large n, a contradiction. Thus a single finite constant exists for the entire class. □

Proof of Theorem 1.3(ii). If (8) failed, there would be $T_n \to \infty$ and $P_n \in \mathcal{E}_{N,\beta}(T_n)$ such that $\|WP_n\|_2 = 1$ and the low-band Fourier norm remained bounded below by a positive constant. Extract the clusters of Lemma 4.1. Each $t_{C,n}$ is the real part of an original frequency, so $|t_{C,n}| \ge T_n \to \infty$.

For $|t_{C,n}| > R$, Fourier translation and Plancherel imply

$$\begin{aligned} \|\mathbf{1}_{[-R,R]} \widehat{e^{it_{C,n}x} A_{C,n}}\|_2^2 &= \int_{|t|\le R} |\widehat{A}_{C,n}(t - t_{C,n})|^2 \, dt \\ &\le \frac{\|A'_{C,n}\|_2^2}{(|t_{C,n}| - R)^2} \le \frac{L_C^2 \|A_{C,n}\|_2^2}{(|t_{C,n}| - R)^2}. \end{aligned} \tag{23}$$

Equation (22) bounds every envelope norm by $\sqrt{2}$. There are finitely many clusters, so the triangle inequality makes the low-band norm of their sum tend to zero, a contradiction. □

Proof of Theorem 1.3(iii). Let C be the constant in (7) and put $\varepsilon(T;R) = \varepsilon_{N,\beta,W}(T;R)$. For every fixed R, Lemma 2.1 and the bounded perturbation estimate give

$$\mathfrak{q}_D[WP] \ge \left\{ \frac{1}{2} \log(1 + 4R^2)(1 - \varepsilon(T;R)^2) - \|D\|C \right\} \|WP\|_2^2 \qquad (P \in \mathcal{E}_{N,\beta}(T)). \tag{24}$$

All Gamma frequencies remain in the original form; its nonnegative low-frequency part was only discarded in this lower estimate. Given $L > 0$, first choose R so that $\frac{1}{4}\log(1+4R^2) - \|D\|C \geq L$. Then choose T large enough that $\varepsilon(T;R)^2 \leq 1/2$. This proves the first divergence in (9). For a prescribed weighted coefficient $L_W > 0$, apply that result with $L = L_W C$ and use (7) to get $\mathfrak{q}_D[WP] \geq L_W \|WP\|_W^2$. □

The order of choices is important: first fix a Fourier band R, then send the minimum real carrier height T to infinity. The extraction proof does not yield a logarithmic-in-T rate for arbitrary configurations.

6 The theta weight and complete prime correction

We now give a concrete realization, including all prime powers. It is constructed directly on $\mathcal{H}_W$; no identification with another energy space, zero-sum formula, or statement about zeta zeros is used.

For $x \geq 0$ define

$$\kappa(x) = e^{x/2} \sum_{j=1}^{\infty} (4\tau_j(x)^2 - 6\tau_j(x)) e^{-\tau_j(x)}, \qquad \tau_j(x) = \pi j^2 e^{2x}, \tag{25}$$

extend evenly, and set

$$W(x) = \frac{\kappa(x)}{2\cosh(x/2)}. \tag{26}$$

Lemma 6.1 (Theta estimates)**.** *The function W in* (26) *is positive, even, smooth, and admissible. For*

$$C_\kappa = 4\pi^2 \sum_{j=1}^{\infty} j^4 e^{-\pi(j^2-1)} < \infty$$

one has

$$0 < W(x) \leq C_\kappa \exp(4|x| - \pi e^{2|x|}). \tag{27}$$

Every fixed derivative of W decreases faster than every exponential.

Proof. The series in (25) and its derivatives converge uniformly on compact subsets of $[0,\infty)$. Every summand is positive, since $\tau_j \geq \pi > 3/2$. To justify smooth evenness at zero, define on all of $\mathbb{R}$

$$F(x) = e^{x/2} \sum_{j=1}^{\infty} e^{-\pi j^2 e^{2x}}.$$

This is smooth by locally uniform differentiated convergence. The Gaussian theta transformation

$$\Theta(t) := \sum_{j\in\mathbb{Z}} e^{-\pi j^2 t} = t^{-1/2}\Theta(1/t) \qquad (t > 0) \tag{28}$$

gives $F(-x) = F(x) + \sinh(x/2)$. For completeness, (28) follows by periodizing $y \mapsto e^{-\pi t y^2}$: its kth Fourier coefficient is $t^{-1/2}e^{-\pi k^2/t}$, by the Gaussian Fourier integral. Both the periodization and this Fourier series converge absolutely and uniformly; evaluating at zero proves the identity. This is the usual Jacobi transformation [1, Eq. 20.7.32].

Apply $\partial_x^2 - 1/4$ to the identity for F. Since it annihilates $\sinh(x/2)$ and commutes with reflection, $F'' - F/4$ is even. Direct differentiation gives precisely (25) on $x \geq 0$. Thus the even extension is smooth.

For $x \geq 0$, dropping the negative term yields

$$\kappa(x) \leq 4\pi^2 e^{9x/2} \sum_{j \geq 1} j^4 e^{-\pi j^2 e^{2x}} \leq C_\kappa e^{9x/2 - \pi e^{2x}}.$$

Since $2\cosh(x/2) \geq e^{x/2}$, this proves (27). After r derivatives, each summand of κ is $e^{x/2}$ times a polynomial of degree at most $r+2$ in τ_j, times $e^{-\tau_j}$. Consequently, for a finite constant C_r and $x \geq 0$,

$$|\kappa^{(r)}(x)| \leq C_r e^{(9/2+2r)x - \pi e^{2x}}.$$

Here one bounds the remaining sum by a constant multiple of $\sum_{j \geq 1} j^{2r+4} e^{-\pi(j^2-1)}$. Derivatives of $(2\cosh(x/2))^{-1}$ are bounded by constants times $e^{-x/2}$ on this half-line. Leibniz's rule, then evenness and smoothness on a compact interval, proves the derivative assertion and admissibility. □

Write $\Lambda(n) = \log p$ if $n = p^k$ with p prime and $k \geq 1$, and $\Lambda(n) = 0$ otherwise. Put

$$a_n = \frac{\Lambda(n)}{\sqrt{n}}, \qquad s_n = \log n, \qquad c_\Gamma = \log \pi - \psi(1/4). \tag{29}$$

Only the elementary bound $0 \leq \Lambda(n) \leq \log n$ will be used.

Proposition 6.2 (Norm absolute convergence of all prime powers)**.** *For $s \geq 0$, the operator*

$$(K_s u)(x) = W(x) u(x+s)$$

is bounded on $\mathcal{H}_W$, with

$$\|K_s\| \leq b(s) := \sup_x \sqrt{W(x) W(x+s)}. \tag{30}$$

The series

$$D_\theta = c_\Gamma M_W + \sum_{n=2}^{\infty} a_n (K_{s_n} + K_{s_n}^*) \tag{31}$$

converges absolutely in operator norm and defines a bounded self-adjoint operator on $\mathcal{H}_W$. Here $M_W u = Wu$. Its form is exactly

$$\langle D_\theta u, v \rangle_W = c_\Gamma \langle u, v \rangle_2 + \sum_{n=2}^{\infty} a_n \left\{ \int_{\mathbb{R}} u(x+s_n) \overline{v(x)}\, dx + \int_{\mathbb{R}} u(x) \overline{v(x+s_n)}\, dx \right\}. \tag{32}$$

The complete series in (32) *is absolutely convergent for all $u, v \in \mathcal{H}_W$.*

Proof. With $y = x + s$,

$$\|K_s u\|_W^2 = \int_{\mathbb{R}} W(y-s) |u(y)|^2\, dy \leq b(s)^2 \|u\|_W^2,$$

which proves (30). The adjoint is $(K_s^* u)(x) = W(x) u(x-s)$, as a change of variables in the weighted inner product shows. Equivalently, weighted Cauchy–Schwarz gives

$$\left| \int_{\mathbb{R}} u(x+s) \overline{v(x)}\, dx \right| \leq b(s) \|u\|_W \|v\|_W. \tag{33}$$

At least one of $|x|$ and $|x+s|$ is at least $s/2$. The function $r \mapsto e^{4r - \pi e^{2r}}$ is decreasing for $r \geq 0$, since its logarithmic derivative is $4 - 2\pi e^{2r} < 0$. Using (27) for that endpoint and $\|W\|_\infty$ for the other gives

$$b(s) \leq (\|W\|_\infty C_\kappa)^{1/2} \exp\left(s - \frac{\pi}{2} e^s \right) \qquad (s \geq 0). \tag{34}$$

Hence

$$\sum_{n=2}^{\infty} a_n b(\log n) \le (\|W\|_\infty C_\kappa)^{1/2} \sum_{n=2}^{\infty} (\log n)\sqrt{n}\, e^{-\pi n/2} < \infty. \tag{35}$$

The last series converges, for example by the ratio test. Also $\|M_W\| \le \|W\|_\infty$. Thus (31) converges absolutely in operator norm, with

$$\|D_\theta\| \le |c_\Gamma| \|W\|_\infty + 2 \sum_{n=2}^{\infty} a_n b(\log n) < \infty. \tag{36}$$

Each partial sum is self-adjoint, so the limit is self-adjoint. The adjoint formula and (33) prove (32) and its absolute convergence. □

No enumeration of primes, truncation of the prime series, or ordinary L^2 boundedness of the complete prime operator is needed. In this weighted setting the prime correction is an actual norm-convergent operator series. Reflection interchanges K_s and K_s^* and commutes with M_W. Thus D_θ preserves the even subspace. The Gamma form also preserves the even/odd decomposition because its multiplier is even.

Corollary 6.3 (Complete theta and prime form)**.** *For the weight* (26)*, define* $\mathfrak{q}_\theta = \mathfrak{t}_\Gamma - \langle D_\theta \cdot, \cdot \rangle_W$*. For each fixed* N *and* β*, both Rayleigh quotients in* (9)*, with* $D = D_\theta$*, tend uniformly to* $+\infty$*. The same statement holds when the class is restricted to even functions. Every prime power and the exact Gamma normalization are retained.*

Proof. Apply Theorem 1.3 using Lemma 6.1 and Proposition 6.2. Restricting the class cannot lower the infimum. □

7 Confluent cosine families and finite matrices

Let W be even and admissible. For $w \in \mathbb{C}$ and $k \ge 0$, put

$$J_{w,k}(x) = W(x)\, \partial_z^k \cos(zx)\Big|_{z=w}. \tag{37}$$

These are even functions; here the word jet means only a parameter derivative and does not impose a zero equation. Suppose that $w_1, \dots, w_r$ are distinct, $\operatorname{Re} w_j \ge T > 0$, $|\operatorname{Im} w_j| \le \beta$, and $m_j \ge 1$. Set $d = \sum_j m_j$ and

$$u = \sum_{j=1}^{r} \sum_{k=0}^{m_j - 1} b_{j,k} J_{w_j,k}. \tag{38}$$

The identity

$$\partial_z^k \cos(zx) = \frac{1}{2}\big((ix)^k e^{izx} + (-ix)^k e^{-izx}\big) \tag{39}$$

shows that $u/W \in \mathcal{E}_{2d,\beta}(T)$.

Corollary 7.1 (Uniform finite cosine matrices)**.** *Fix* N*,* β*, and a bounded self-adjoint* D *on* $\mathcal{H}_W$*. For every* $L > 0$ *there is* $T_N(L) < \infty$ *such that every combination* (38) *with* $d \le N$ *and* $\min_j \operatorname{Re} w_j \ge T_N(L)$ *satisfies*

$$\mathfrak{q}_D[u] \ge L \|u\|_W^2. \tag{40}$$

For the nonredundant prefix list displayed above, the form matrix is positive definite at those heights. More precisely it dominates L *times the weighted Gram matrix as a Hermitian form on the coefficient space.*

Proof. The inequality is Theorem 1.3 at complexity $2N$. To justify strictness of the coefficient form, recall exponential-polynomial independence. If $\sum_j p_j(x)e^{i\lambda_j x} = 0$ with distinct λ_j, apply $\prod_{h\neq j}(\partial_x - i\lambda_h)^{\deg p_h+1}$. Every term except the jth vanishes. After removing $e^{i\lambda_j x}$, the remaining operator on p_j is a product of powers of $\partial_x + i(\lambda_j - \lambda_h)$. Each factor is invertible on the finite polynomial space: in the monomial basis its matrix is triangular with a nonzero constant diagonal. Therefore $p_j = 0$ for every j.

The frequencies $w_1, \dots, w_r, -w_1, \dots, -w_r$ are all distinct because the first group has positive real parts and the w_j are distinct. Equation (39) and independence imply that the prefix jets are linearly independent. Hence a nonzero coefficient vector represents a nonzero u, and (40) gives $\mathfrak{q}_D[u] > 0$. □

For repeated or redundant columns the same inequality holds, but the coefficient matrix can be singular. For a sparse selection at w_j, the correct complexity cost is one plus the largest derivative order selected, not the number of selected columns. Near collisions the unnormalized Gram matrix can be ill-conditioned; no uniform lower eigenvalue in raw Euclidean coordinates is asserted.

This corollary applies to arbitrary frequencies meeting its hypotheses. It neither uses nor proves that a given frequency is a zero of a special function. In particular, no conclusion about all actual zeta-zero jets, a centered half inequality, or the Riemann hypothesis follows from it.

8 A fixed width cosine packet

Although the general theorem has no stated rate, a common bounded offset window permits a useful explicit form of estimate with unevaluated constants. Fix $R_0 \geq 0$ and $\beta \geq 0$. Let P_+ have complexity at most N and frequencies in $|\operatorname{Re}\lambda| \leq R_0$, $|\operatorname{Im}\lambda| \leq \beta$. Put $A = WP_+$ and

$$u_T(x) = \frac{1}{2}(e^{iTx}A(x) + e^{-iTx}A(-x)), \qquad T > 0, \tag{41}$$

where W is even. By Lemma 3.1, choose a constant $C \geq 1$ depending only on N, R_0, β, W such that $\|A\|_W, \|A'\|_2 \leq C\|A\|_2$.

Proposition 8.1 (Carrier estimate)**.** *For $T \geq 2C$,*

$$\mathfrak{q}_D[u_T] \geq \left\{\frac{1}{2}\log(1+T^2)\left(1 - \frac{16C^2}{T^2}\right) - 4\|D\|C^2\right\}\|u_T\|_2^2. \tag{42}$$

Proof. Integration by parts gives

$$\left|\int_{\mathbb{R}} e^{2iTx}A(x)\overline{A(-x)}\,dx\right| \leq \frac{\|A'\|_2\|A\|_2}{T} \leq \frac{C}{T}\|A\|_2^2.$$

Expanding (41) therefore yields $\|u_T\|_2^2 \geq \|A\|_2^2/4$ for $T \geq 2C$. The weighted triangle inequality gives $\|u_T\|_W^2 \leq C^2\|A\|_2^2 \leq 4C^2\|u_T\|_2^2$. Using (23) with the two carriers $\pm T$ and band $[-T/2, T/2]$, including their factors $1/2$, gives

$$\|\mathbf{1}_{[-T/2,T/2]}\widehat{u_T}\|_2 \leq \frac{2\|A'\|_2}{T} \leq \frac{4C}{T}\|u_T\|_2.$$

Now apply (11) outside that band and subtract $\|D\|\|u_T\|_W^2$. □

The right side tends to a positive and then arbitrarily large coefficient as $T \to \infty$ with the window and complexity fixed. The constants have not been evaluated, so this is not an effective numerical threshold. If the displayed lower bound is negative at a smaller carrier, that fact is not a negative value of the form.

9 Why complexity cannot be removed

The quantifiers in Theorem 1.3 have the order

$$\text{for each fixed } N, \beta, L \text{ there is a height } T(N, \beta, L).$$

They do not provide a single height for all complexities. Taking the span of all high-frequency jets allows unbounded complexity and is not covered by the theorem. The following constructions demonstrate that two key uniform estimates genuinely fail on the larger arbitrary-frequency class.

Proposition 9.1 (Failure of complexity independent norm comparison)**.** *For any admissible W, let $P_m(x) = x^{2m}$. Then*

$$\frac{\|WP_m\|_W^2}{\|WP_m\|_2^2} \longrightarrow \infty. \tag{43}$$

Proof. Define the probability measure

$$d\nu_m(x) = \frac{W(x)^2 x^{4m}\,dx}{\int_{\mathbb{R}} W(y)^2 y^{4m}\,dy}.$$

For each fixed $R \geq 1$,

$$\nu_m([-R,R]) \leq \left(\frac{R}{R+1}\right)^{4m} \frac{\int_{-R}^{R} W(x)^2\,dx}{\int_{R+1}^{R+2} W(x)^2\,dx} \longrightarrow 0.$$

The denominator in the fixed ratio is strictly positive. Since $W(x) \to 0$ as $|x| \to \infty$, given $B > 0$ choose R so that $1/W \geq B$ outside $[-R, R]$. Then

$$\frac{\|WP_m\|_W^2}{\|WP_m\|_2^2} = \int_{\mathbb{R}} \frac{1}{W}\,d\nu_m \geq B(1 - \nu_m([-R,R])).$$

Take the lower limit and then let $B \to \infty$. □

Proposition 9.2 (Two high carrier locations with growing degree)**.** *Suppose W is even and admissible. There exist $T_n \to \infty$ and real even exponential polynomials P_n whose only frequency locations are $\pm T_n$, but whose complexities are unbounded, such that*

$$WP_n \longrightarrow W \quad \text{in } \mathcal{H}_W \text{ and in } H^1(\mathbb{R}), \quad \text{hence in } \mathcal{X}_W. \tag{44}$$

Consequently uniform low-frequency escape fails when no complexity bound is imposed. For every fixed bounded self-adjoint D,

$$\frac{\mathfrak{q}_D[WP_n]}{\|WP_n\|_2^2} \longrightarrow \frac{\mathfrak{q}_D[W]}{\|W\|_2^2} < \infty. \tag{45}$$

Proof. For fixed $T > 0$, put

$$S_{m,T}(x) = \sum_{k=0}^{m} \frac{(-iTx)^k}{k!}, \qquad P_{m,T}(x) = \frac{1}{2}\big(e^{iTx} S_{m,T}(x) + e^{-iTx} S_{m,T}(-x)\big).$$

These functions are real and even on $\mathbb{R}$, have frequencies $\pm T$, and have complexity $2(m+1)$. For fixed T, the Taylor series and its derivative converge locally uniformly, so $P_{m,T} \to 1$ and $P'_{m,T} \to 0$ pointwise. For all m,

$$|S_{m,T}(x)| \leq e^{T|x|}, \qquad |S'_{m,T}(x)| \leq T e^{T|x|},$$

and hence $|P_{m,T}| \le e^{T|x|}$ and $|P'_{m,T}| \le 2Te^{T|x|}$. Equation (2) supplies integrable dominating functions for the weighted norm, ordinary norm, and derivative of $W(P_{m,T}-1)$. Dominated convergence proves $WP_{m,T} \to W$ in $\mathcal{H}_W \cap H^1$ at fixed T.

Choose any $T_n \to \infty$ and then choose m_n so large that the sum of the two norm distances is below $1/n$. Set $P_n = P_{m_n,T_n}$. Equation (12) proves (44) in $\mathcal{X}_W$. Because $W \in L^1$, $\widehat{W}$ is continuous and $\widehat{W}(0) = (2\pi)^{-1/2}\int W > 0$. Thus some fixed low Fourier band has positive limiting mass, contradicting any complexity independent escape statement. Finally $\mathfrak{q}_D$ is continuous on $\mathcal{X}_W$: Cauchy–Schwarz holds for $\mathfrak{t}_\Gamma$, and D is bounded. This proves (45). The complexities must be unbounded, otherwise Theorem 1.3(ii) would contradict the nonzero limiting low-band mass. □

These are auxiliary arbitrary-frequency examples. The first has frequency zero; the second prescribes arbitrarily high polynomial order at two chosen frequencies. Neither is asserted to use actual zeta zeros or their actual multiplicities. Neither proves that $\mathfrak{q}_\theta$ is negative at any vector, and (45) does not determine the sign of $\mathfrak{q}_\theta[W]$. They rule out deleting the complexity hypothesis from this general mechanism, not a possible theorem using additional zero-specific structure.

A An entire two frequency formula

For the theta weight and D_θ of Section 6, define

$$e_z(x) = W(x)\cos(zx), \qquad H_s(t) = \int_{\mathbb{R}} W(x+s/2)W(x-s/2)\cos(tx)\,dx \quad (s \ge 0). \tag{46}$$

The following exact formula records the complete form on cosine vectors. It is not a zero-sum representation and makes no zero-location assumption.

Proposition A.1 (Entire bilinear kernel)**.** *The function $\mathcal{K}(z,v) = \mathfrak{q}_\theta(e_z, e_{\overline{v}})$ is entire on $\mathbb{C}^2$ and is symmetric in z, v, even in each variable, and real under simultaneous complex conjugation. It satisfies*

$$\begin{aligned}\mathcal{K}(z,v) = \int_0^\infty \rho(s)\Big[&H_0(z-v) + H_0(z+v) \\ &- \cos\big((z+v)s/2\big)H_s(z-v) - \cos\big((z-v)s/2\big)H_s(z+v)\Big]\,ds \\ &- \frac{c_\Gamma}{2}\big[H_0(z-v) + H_0(z+v)\big] \\ &- \sum_{n=2}^{\infty} a_n\Big[\cos\big((z+v)s_n/2\big)H_{s_n}(z-v) + \cos\big((z-v)s_n/2\big)H_{s_n}(z+v)\Big].\end{aligned} \tag{47}$$

The Gamma bracket is kept combined. The integral and series converge locally uniformly, and every fixed parameter derivative can be taken under them. In particular,

$$\mathfrak{q}_\theta(J_{w,k}, J_{v,\ell}) = \partial_z^k \partial_\eta^\ell \mathcal{K}(z,\eta)\Big|_{z=w,\,\eta=\overline{v}}. \tag{48}$$

Proof. For a compact parameter set, e_z and each of its fixed parameter derivatives have uniformly bounded $\mathcal{H}_W$ and H^1 norms by the theta tails. The same estimates, with one additional derivative, justify complex differentiation by dominated convergence in both norms. Thus $z \mapsto e_z$ is entire into $\mathcal{X}_W$ by (12).

Set $C_s(z,v) = \int_{\mathbb{R}} e_z(x+s)e_v(x)\,dx$. With $y = x + s/2$, the weight product is even in y. The product-to-sum identity, whose odd sine terms integrate to zero, yields

$$C_s(z,v) = \frac{1}{2}\Big[\cos\big((z+v)s/2\big)H_s(z-v) + \cos\big((z-v)s/2\big)H_s(z+v)\Big]. \tag{49}$$

In particular C_s is symmetric in z, v. Translation invariance gives

$$\int_{\mathbb{R}} (e_z(x+s) - e_z(x))(e_v(x+s) - e_v(x))\, dx = 2C_0(z,v) - 2C_s(z,v).$$

The two prime orientations also both equal $C_{s_n}(z,v)$. Substitution into the exact forms proves (47).

On compact parameter sets, the Gamma bracket is $O(s^2)$ near zero, bounded by $s^2\|e_z'\|_2\|e_v'\|_2$; for large s it is bounded by $4\|e_z\|_2\|e_v\|_2$. Both bounds are integrable against $\rho(s)\, ds$. They remain valid for fixed parameter derivatives, with the corresponding differentiated vectors. For the prime part, (33) and (35) give absolute locally uniform convergence, also after every fixed parameter derivative. Each H_s is entire by the same theta domination. This proves all convergence and analyticity assertions. Evenness, symmetry and the real conjugation identity follow from the formulas; differentiation gives (48). Notice that the sesquilinear kernel is $\mathfrak{q}_\theta(e_z, e_v) = \mathcal{K}(z, \overline{v})$, not $\mathcal{K}(z,v)$. □

Scope and status

The proved conclusion is uniform coercivity for each fixed finite complexity and strip, including collisions and arbitrary real separations. The full-prime specialization is justified by an operator-norm summability proof in the weighted space. No positivity at every height, no complexity-independent threshold, no coefficient-conditioned eigenvalue bound, and no positivity theorem for all actual zeta-zero jets is claimed. The manuscript does not invoke a physical energy bridge or a Guinand zero-sum identity. No numerical research, prime or zero enumeration, matrix computation, or numerical sign certificate is used. This is a research manuscript; it is not a claim of peer-reviewed publication or formal proof-assistant verification.

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